Engineering - Graduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Calculate the Laplace transform oft^nu(t).
   (a) L{t^nu(t)} = [n! / (s^(n+3))]
   (b) L{t^nu(t)} = [(n+1)! / (s^(n+1))]
   (c) L{t^nu(t)} = [n! / (s^n)]
   (d) L{t^nu(t)} = [n! / (s^n+ 2)]
   (e) L{t^nu(t)} = [n! / (s^n+ 1)]
2. A response transformV_ab(s) is related to a source transform V(s) by V_ab(s) = [{3V(s) + 3s + 27} / (s^2 + 6s + 8)] Findv_ab(t) if v(t) = 16e^-3tu(t).
   (a) - 48e^-3t - (68 / 2)e^-2t + (33 / 2)e^-4t
   (b) 16e^-3t + (69 / 2)e^-2t + (33 / 2)e^-4t
   (c) - 48e^-3t+ (69 / 2)e^-2t + (33 / 3)e^-4t
   (d) - 48e^-3t+ (70 / 2)e^-2t + (33 / 2)e^-4t
   (e) - 49e^-3t + (69 / 2)e^-2t + (33 / 2)e^-4t
3. Find the attenuation per meter along a wave guide for an appliedwave length λ_0 = 2 meters, if the cutoff wave length ofthe guide isλ_oc= 20 cm.
   (a) 400 db/meter
   (b) 545 db/meter
   (c) 600 db/meter
   (d) 325 db/meter
   (e) 250 db/meter
4. Compute the impact stress in a steel rod which is hit by a block weighing 45 lbs. The speed of the block is 40 in./sec. The rod is 18 in. long and has a diameter of 1(1 / 4) in.
   (a) 8,762.3 psi
   (b) 4,321.0 psi
   (c) 25,678.9 psi
   (d) 6.25 lbs
   (e) 30,000 psi
5. In decimal number system what is MSD

(a) First two digits from right to left
(b) Median of all digits
(c) First digit from right to left
(d) First digit from left to right
(e) Mode of all digits

True / False (10 marks)

Answer True or False

6. True or False: Silicon is regarded as a highly detrimental impurity in magnetic materials and signi
7. True or False: The theoretical variance of the given probability distribution is 1.125.
8. True or False: To drive a load at 1750 r/min with a torque of 60 lb.ft, a horsepower of 20 hp is rec
9. True or False: A single phase full bridge inverter can operate in load commutation mode when th
10. True or False: The power delivered by a roller chain of 3/4 in. pitch on a 34-tooth sprocket at 500

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of temperature measurement in a chemistry laboratory, specifically conv implications for experimental accuracy.
12. Given the current $i_1(t) = 2e^{lve.}$ A flowing into the positive terminal of a circuit element, analyze the p $10s$ using the provided voltage equations. Explain your reasoning and calculations in detail.

End of Paper